

Name: _____

Class: 2ma2____ / 2IPma2____



NATIONAL JUNIOR COLLEGE
2009 SH2 Preliminary Examination

MATHEMATICS
Higher 2

9740/02
Paper 2

Additional Materials: Answer Paper
List of Formulae (MF15)
Cover Sheet

3 hours

16 September 2009

0815 – 1115 hours

INSTRUCTIONS TO CANDIDATES

Write your name, registration number, subject tutorial group, on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **ALL** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise. Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in the brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This question paper consists of 10 printed pages.

Section A: Pure Mathematics [40 marks]

- 1** Given that the second term in the expansion of $\left(kx + \frac{1}{x}\right)^{-\frac{1}{3}}$ in ascending powers of x is $-3x^{\frac{7}{3}}$, find the value of constant k .

State the range of values of x for which the series expansion of $(1 - 4x)^8 \left(kx + \frac{1}{x}\right)^{-\frac{1}{3}}$

is valid.

[4]

- 2** A Sumo wrestler would like to have fish fillet, salad and fries for breakfast. As he is on a special diet, he must make sure that his intake (in grams) of protein, carbohydrates and fats per meal is in the ratio of 4:8:3. The table below shows the nutritional breakdown for one serving of each item.

	Protein (in grams)	Carbohydrates (in grams)	Fats (in grams)
Fish Fillet	150	60	25
Salad with dressing	15	30	5
Fries	5	250	110

Calculate the ratio of the servings of fish fillet, salad and fries that the wrestler should take.

[4]

- 3** Given that $y = e^x \sin x \cos x$, prove that $\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 5y$.

By further differentiation of this result, find the Maclaurin's series for y , up to and including the term in x^3 .

[5]

Deduce the Maclaurin's series for the function $e^x \cos 2x$ as far as the term in x^2 .

[2]

4 (a) The sequence of numbers u_n is given by $u_1 = 4$, $u_n = \frac{(n+1)^2}{n} u_{n-1}$, $n \in \mathbb{N}^+$, $n > 1$.

(i) Find u_2 , u_3 and u_4 . [2]

(ii) By considering $\frac{u_n}{n!}$ or otherwise, write down a conjecture for u_n in terms of n .

Use the method of mathematical induction to prove your conjecture. [5]

(b) The sequence of real numbers $x_1, x_2, x_3, \dots$ satisfies the recurrence relation

$$x_{n+1} = \frac{1}{2}(e^{x_n} - 3).$$

(i) The equation $\frac{1}{2}(e^x - 3) - x = 0$ has two real roots denoted by a and b where

$a < b$. Find a and b , each correct to 3 decimal places. [2]

(ii) By considering $x_{n+1} - x_n$, show that

$$a < x_n < b \Rightarrow x_n > x_{n+1}$$

and $x_n < a$ or $x_n > b \Rightarrow x_n < x_{n+1}$. [2]

(iii) Describe the behaviour of the sequence when $x_1 = 1$ and $x_1 = 3$. [2]

5 (a) The curve C is defined by the parametric equations

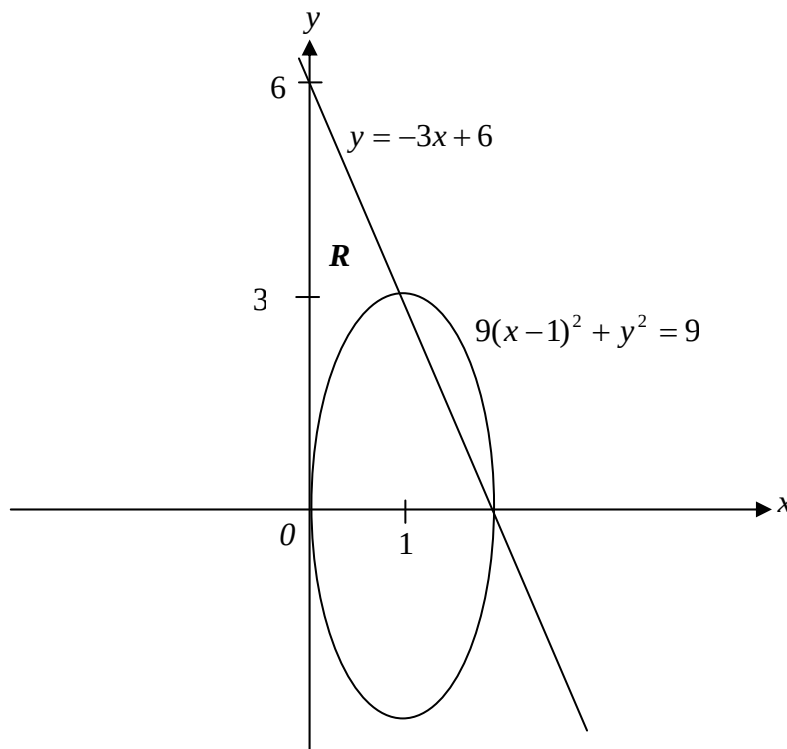
$$x = t^2 + 2t, \quad y = t^2 - t, \quad \text{where } t \in \mathbb{R}.$$

- (i) Show algebraically that $y \geq -\frac{1}{4}$ for all values of t . [1]
- (ii) Sketch the curve C , indicating clearly the axial intercepts. [2]
- (iii) Find the exact area of the region bounded by the curve C and the x -axis from $x = 0$ to $x = 8$. [4]

(b) The region R is bounded by the curve $9(x-1)^2 + y^2 = 9$, the line $y = -3x + 6$ and the y -axis shown in the diagram below.

Find the coordinates of the points of intersection between $9(x-1)^2 + y^2 = 9$ and $y = -3x + 6$. [2]

Find the numerical value of the volume of revolution formed when R is rotated completely through 4 right angles about the y -axis. [3]



Section B: Statistics [60 marks]

- 6 In order to find out the percentage of students who will still travel overseas despite the H1N1 outbreak, a group of four undergraduates decide to sample 100 students. During allocation of task, the four undergraduates A, B, C, D decided that each of them will sample 25 students, and taking their samples from a junior college, a polytechnic, a secondary and a primary school respectively.

- (i) Name the sampling model chosen by these four undergraduates. [1]
- (ii) Identify a significant area of weakness in their above proposed model and suggest an improvement to the model and explain why that will address the weakness. [3]

- 7 In year 2008, a survey was done by the Ministry of Manpower (MOM) to determine the amount of daily wages, denoted as X , that part-time workers were receiving. Past records show that on average, the wages that each worker received was \$30. A year later, MOM wants to find out if there is a change in the mean wage. Data from 90 workers are collected and their wages are summarized as shown below:

$$\bar{x} = 31, \quad \sum (x - 31)^2 = 1947.$$

Calculate the unbiased estimate of the variance of X . [1]

MOM decides to do a test on the collected data. Find the range of values of α if MOM were to conclude that there is a change in the mean wage at α % level of significance. [3]

An audit company suspects that the mean wage of the part-time workers per day at a particular restaurant is at most \$25. The wages of six part-time workers are \$21.50, \$24, \$25.50, \$27, \$28 and \$32. Conduct an appropriate test at 10% level of significance.

State an assumption made for the test to be valid. [5]

- 8** A set of eight cards is marked with a letter each to make up the word TOMATOES.
- (a) Four of these cards are selected at random. Find the number of selections if the cards are drawn without replacement. [3]
- (b) Find the number of arrangements of all the eight cards
- (i) in a row, such that no vowels are adjacent to each other. [2]
- (ii) in two distinct rows of four cards such that no identical letters are in the same row. [2]
- 9** It is found that everyday for any clinic, on average 2 patients are diagnosed with flu. It may be assumed that the flu cases are independent.
- (i) Find the probability that no less than 3 patients are diagnosed with flu at a clinic on any day. [2]
- (ii) Fifty clinics are randomly chosen, what is the probability that on a randomly chosen day, the mean number of patients who are diagnosed with flu per clinic is between 2 and 5? [3]
- It is also known that 93% of patients who are diagnosed with flu are given special medical leave. Sixty independent patients diagnosed with flu are examined.
- Using a suitable approximation, find the probability that more than 57 of the examined patients diagnosed with flu were given special medical leave. [3]

- 10** Seven balls of which three are white and four are black are in a box. Three balls are randomly picked and put in bag A .

Let X be the random variable that denotes the number of white balls in bag A .

Copy and complete the following probability distribution table for X . [3]

X	0	1	2	3
$P(X = x)$	$\frac{4}{35}$			$\frac{1}{35}$

- (i) Find the probability that there are exactly three white balls in bag A , given that at least one white ball has been drawn and put in bag A . [2]

Suppose it is now known that bag A has exactly two white balls and one black ball and that all the remaining balls are then put in a bag B .

- (ii) A bag is selected at random and one ball is drawn from it. Find the probability that the ball is white. [2]
- (iii) A ball is drawn repeatedly from bag B with replacement until a white ball is drawn. Let Y denote the number of draws (including the last draw) that must be made until a white ball is drawn. Calculate the least value of n such that $P(Y \leq n) > 0.99$. [3]

- 11** At a clinical laboratory, a machine is used to measure the growth of a certain bacteria at fixed time intervals and the results are tabulated as follow:

Time (days), x	5	10	24	35	48	55	67
Number of bacteria (thousands), y	1.3	42.0	14.8	30.1	60.8	81.3	98.6

It was discovered that one of the results may be wrong. Identify the result that is most likely to be incorrect. Justify your answer. [2]

The incorrect result which you identified above is rectified. The data with the correct result yields the following regression lines y on x and x on y respectively,

$$y = -14.16830 + 1.61302x \quad \text{and} \quad x = 9.97265 + 0.59168y.$$

- (i) Determine the value of the correct result, correct to 1 decimal place. [3]
- (ii) State, giving a reason, which of the least squares regression lines, y on x or x on y , should be used to express a possible linear relation between y and x . [1]
- (iii) Using the regression line you chosen in (ii), an estimate for the number of bacteria in 40 days is obtained. Comment on its reliability. [2]
- (iv) For each of the seven sample values of x , Y' is given by $Y' = a + bx$, where a and b are any real constants. Explain why $\sum (y - Y')^2 \geq c$ where c is a constant to be determined. [2]

- 12** On Crabby Island, a fishmonger rears many types of live seafood in his kelong, one of which is a particular type of crab named the Hairy Crab. The weight H , in kg, of a random Hairy Crab follows a normal distribution with mean μ kg and standard deviation σ kg.

Given that $P(H < 1.5) = P(H > 3.5) = 0.33$, state the value of μ . [1]

(i) Show that $\sigma = 2.27$. [2]

(ii) Find the probability that out of four randomly chosen crabs, two weigh between 1.5 kg and 3.5 kg each and the other two weigh more than 3.5 kg each. [3]

(iii) Using a suitable approximation, find the probability that, out of 100 Hairy Crabs, at least 80 will weigh more than 1.5 kg. [3]

Another type of crab named Flower Crab found on Crabby Island follows an independent normal distribution with mean weight 3.2 kg and standard deviation 1.3 kg. Find the largest value of a such that the probability that the average weight of two randomly chosen Hairy Crabs and three randomly chosen Flower Crabs exceeds a kg is more than 0.6. [3]

End of Paper



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INSTRUCTIONS TO CANDIDATES

Write your name, registration number, subject tutorial group and subject tutor's name in the spaces provided on the cover sheet and attach it on top of your answer paper.

Circle the questions you have attempted and arrange your answers in **NUMERICAL ORDER**.

Name: _____ Registration Number: _____

Subject Tutorial Group: 2ma2____ / 2IPma2____ Subject Tutor: _____

Question	Marks	Max. Marks	Question	Marks	Max. Marks
1		4	7		9
2		4	8		7
3		7	9		8
4		13	10		10
5		12	11		10
6		4	12		12
Presentation: -1 / -2					
Total		100	Grade		